

NMR Probe Double-Tuned Matching Boxes for Double-Resonance FT Experiments on Hydrogen and Fluorine

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Commercially available transmitter double-tuned matchboxes for $^1\text{H}\{^{19}\text{F}\}$ or $^{19}\text{F}\{^1\text{H}\}$ double-resonance experiments are of limited utility for pulsed NMR FT experiments. New double-tuning networks for the Varian V-4412 probe are described for carrying out these experiments at 94 and 100 MHz for ^{19}F and ^1H , respectively. The advantages and performance of these new matchboxes for FT experiments are discussed.

INTRODUCTION

An important class of NMR experiments in our laboratory involves $^1\text{H}\{^{19}\text{F}\}$ or $^{19}\text{F}\{^1\text{H}\}$ double-resonance techniques. However, in a crossed-coil spectrometer the strong coupling between the excitation and decoupling coils makes it difficult to tune them to slightly different frequencies. Varian Associates avoids this problem on the XL-100 by providing transmitter "matchboxes" for the V-4412 probe to double-tune the transmitter coil when observation and decoupling are to be performed at 94 and 100 MHz, the Larmor frequencies of ^{19}F and ^1H , respectively. These matchboxes were apparently designed for cw experiments and provide good coupling efficiency for the H_2 (decoupling) power at the expense of considerably reduced H_1 (observation) efficiency.

The matchbox requirements for pulsed NMR FT experiments are different from the cw requirements; in FT experiments it is often of greater importance to have good coupling efficiency for the H_1 power, and it is desirable to maintain fairly good coupling efficiency for both H_1 and H_2 . An additional requirement is that the H_2 amplifier be protected from the H_1 pulse; this is necessary because the H_1 power is typically more than an order of magnitude larger than that obtainable from the H_2 transmitter.

A set of double-tuned matchboxes designed to meet the criteria for proton-fluorine FT double-resonance experiments is described below. These matchboxes are used as direct replacements for the corresponding Varian matchboxes (part numbers 934729-94 and 934729-99).

EXPERIMENTAL

The single-frequency matchbox design used by Varian to match the transmitter coil L to 50 ohms (resistive) is given in Fig. 1a. Although the match could be made with two rather than three capacitors, the design shown allows the floating capacitor C_2 to be

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fixed and the fine tuning to be accomplished by varying C_1 and C_3 . The circuit shown in Fig. 1b is electrically equivalent at frequency f_1 to that of Fig. 1a when the series combination of C_4 and L_4 is resonant at f_1 , and the parallel combination of C_3' and C_3'' is equal to C_3 . A matching network at the point indicated by the parentheses can be inserted to couple H_2 power at frequency f_2 into the probe; moreover, in this circuit the f_2 transmitter is protected from the H_1 pulse by the C_4L_4 series resonant circuit.

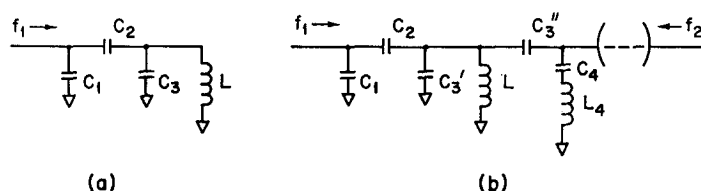


FIG. 1. Matchbox circuits for coupling power into the transmitter coil L . (a) Single-frequency matchbox design. (b) Double-tuned matchbox circuit. The two circuits are equivalent with respect to the f_1 input when $C_3' + C_3'' = C_3$ and the series combination C_4L_4 is resonant at f_1 . The parentheses indicate the position of matching networks to couple f_1 power into the probe.

Calculations using approximate values for the Varian 4412 probe indicated that when f_2 is the higher frequency, the probe circuit appears inductive to the H_2 input and a capacitive matching network is required to couple the H_2 power into the probe. When f_2 is the lower frequency, these calculations indicated that an inductive matching network is required. The actual matching circuits used are shown in Fig. 2, where the series

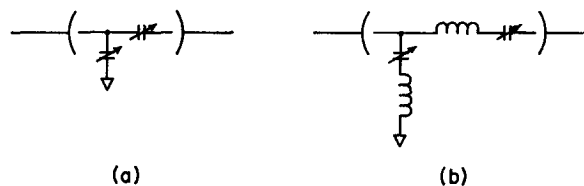


FIG. 2. Matching networks for coupling power at frequency f_2 into the probe. (a) Matching network for the $^{19}\text{F}\{^1\text{H}\}$ double-tuned matchbox. (b) Matching network for the $^1\text{H}\{^{19}\text{F}\}$ double-tuned matchbox. The series LC combinations are used to achieve inductive tuning through the use of variable capacitors.

LC circuits in Fig. 2b are expected to be inductive at frequency f_2 and were used to achieve inductive tuning without the use of magnetic components.²

Although capacitor C_1 in Fig. 1b was initially included in order to allow the capacitance parallel to L as seen by f_2 to be varied, its presence was found to be unnecessary in both matchboxes. In addition, the capacitor C_3' was found to be unnecessary in the $^1\text{H}\{^{19}\text{F}\}$ matchbox; otherwise, the matchbox circuits were constructed as indicated in Figs. 1b and 2. Because the tuning of the matchboxes was accomplished through observation of the voltages induced across L , the actual match of the probe to the H_1 and H_2 transmitters is not known. Some H_1 power dissipation is expected in L_4 , so the

² The use of variable capacitors in the circuit of Fig. 2b could be avoided through utilization of non-magnetic adjustable coils.

efficiency with which H_1 power is coupled into L from the double-tuned matchbox circuit in Fig. 1b is expected to be diminished with respect to the efficiency of the circuit of Fig. 1a. Similarly, the H_2 efficiency is expected to be diminished, not only because of H_2 power dissipation in L_4 , but also to some extent because of H_2 power dissipation in the H_1 transmitter, although in the circuits used the small value of C_2 limits this latter dissipation.

The efficiencies of the double-tuned matchboxes with respect to single-frequency matchboxes are shown in Table I, which lists both voltage and power ratios. Since significant power losses are not expected to occur in the single-frequency matchboxes,

TABLE I
DOUBLE-TUNED MATCHBOX EFFICIENCIES IN TERMS OF rf POWER DELIVERED TO THE TRANSMITTER COIL

Double-tuned matchbox	H_1 efficiency power ^a (voltage)	H_2 efficiency power ^b (voltage)
$^{19}\text{F}\{^1\text{H}\}$	0.79 (0.89)	0.48 (0.69)
$^1\text{H}\{^{19}\text{F}\}$	0.64 (0.80)	0.40 (0.63)

^a H_1 power at frequency f_1 compared to that of a single-frequency matchbox tuned at f_1 .

^b H_2 power at frequency f_2 compared to that of a single-frequency matchbox tuned at f_2 .

the comparison is a measure of the power losses due to the double-tuned matchboxes. As may be seen, in both cases well over half of the H_1 power and nearly half of the H_2 power are preserved.

Johanson 7585 high rf voltage quartz trimmer capacitors (variable) were used for the high-voltage capacitors C_2 , C_3' , C_3'' , and C_4 .³ Unfortunately, these capacitors were found to break down at power levels of 1 kW,⁴ and it was necessary to reduce the power to somewhat less than 500 W to guard against capacitor breakdown. Even at these reduced power levels, the 90° pulse lengths have been between 10 and 15 μsec .

ACKNOWLEDGMENTS

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³ While the Johanson 7585 trimmers were nonmagnetic, the 7168 trimmers, carrying very similar specifications, were found to have magnetic components.

⁴ Our XL-100 is equipped with a Nicolet Technology Corporation 1 kW power amplifier.